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Effects of Social Partners on Pituitary-Adrenal Activity During Novelty Exposure in Adult Female Squirrel Monkeys

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HENNESSY, M. B. *Effects of social partners on pituitary-adrenal activity during novelty exposure in adult female squirrel monkeys.* PHYSIOL BEHAV **38**(6) 803–807, 1986.—Adult female squirrel monkeys from established social groups were exposed to novel surroundings either while alone or while with an adult female partner. The partner was either familiar and judged to be affiliative with the subject, familiar and not judged to be affiliative with the subject, or unfamiliar. The females' plasma cortisol levels were higher at the end of the 25th hr of exposure than at the end of the first hr when the females were alone or with a familiar partner, but not when they were with an unfamiliar partner. In no instance were cortisol concentrations greater when females were tested alone than when they were tested in the presence of a companion. Behavioral signs of distress were greater during the first as compared to the 25th hr of exposure and showed no influence of the presence of a partner. Contact/proximity between monkeys was greater during the 25th than during the first hour of exposure in the familiar partner judged to be affiliative and in the unfamiliar partner conditions. These results contrast with those of studies investigating the disruption of the mother-infant relationship in squirrel monkeys, and provide further evidence that the pituitary-adrenal system of these animals is not responsive to the disruption of adult social relationships.

Social influences Pituitary-adrenal activity Cortisol Novelty exposure Squirrel monkeys

NUMEROUS lines of evidence support the view that the presence or absence of social companions can have major effects on various physiological responses and associated disease processes in humans [1,21]. The loss of significant others is associated with deleterious physiological reactions, and the presence of, or interaction with, these individuals appear to moderate the impact of pathogens and environmental stressors [1,6]. While the relation between social support and well being seems firmly established, we know little of the critical aspects of social relationships nor of their physiological sequelae that could be responsible for such beneficial effects. Recent studies examining the activity of the pituitary-adrenal system following social manipulations in nonhuman primates bear directly on this issue.

Pituitary-adrenal activity increases in a variety of aversive or unpleasant situations, and continued activation of this system is associated with signs of impaired immunological competence [5, 13, 28]. In the squirrel monkey (*Saimiri sciureus*), brief involuntary separation of mother and infant has repeatedly been observed to produce marked increases in plasma concentrations of cortisol in both members of the dyad [3, 23, 27, 31]. The infant's cortisol response does not appear to be due merely to its dependency on the mother or to the novelty of being left alone for the first time, because infants exhibit cortisol elevations even when they are physically independent and have been separated on 80 previous occasions [15]. Rather the plasma cortisol elevations of both animals seem to result from the disruption of a specific social relationship—the attachment of mother and infant.

In contrast, manipulation of other social bonds or relationships in juvenile and adult *Saimiri* has been found to have little or no effect on pituitary-adrenal activity. Adult male, and juvenile male and female *Saimiri* have been found to exhibit no change in plasma cortisol levels when a like-aged peer is removed from the home cage [2, 14, 17]. In other tests, adult males, adult lactating females and juvenile males and females were separated from peers in novel or otherwise threatening environments [2, 14, 17, 18]. In some of these instances, cortisol was significantly elevated over resting levels when the subjects were tested individually, but not when they were tested together with social partners. However, in no case were the cortisol levels of animals tested individually significantly greater than those of animals tested with partners. This relatively weak and somewhat contradictory evidence for an effect of the presence/absence of companions on pituitary-adrenal responsiveness stands in sharp contrast to the clear, reliable and persistent activation of the pituitary-adrenal system that is observed during separation of mothers from their infants [3, 15, 23, 27, 31]. It seems that disruptions of the mother-infant bond elicits a pituitary-

adrenal "stress response" because this relationship is especially strong or intense in *Saimiri*, and disruptions of other social relationships formed between adults or juveniles is not sufficiently disturbing to produce this effect.

However, this conclusion might be premature. Jordan *et al.* [18] found that when two mother-infant pairs of squirrel monkeys were maintained together in a small cage, removal of one pair produced plasma cortisol elevations in both members of the remaining pair. It may be that the close confinement imposed on the dyads in this study made the separation particularly traumatic. On the other hand, the data from these mothers suggest that disruption of the relationship formed between adult females may be of sufficient stimulus intensity to activate the pituitary-adrenal system. This possibility is consistent with the dynamics of squirrel monkey social organization. That is, cliques of females appear to constitute the most cohesive adult subgroup in squirrel monkey troops [10,22].

Therefore, the present study re-examined this issue by measuring plasma cortisol levels in adult females exposed to a novel environment either while alone or while with a familiar or unfamiliar adult female peer. In performing separation studies, one has the choice of isolating the animal to be studied somewhere outside of the home cage, or alternatively, of removing the subject's social companions so that the subject is isolated in the home cage. In the present study, it was decided to perform all testing outside of the home cage both because this alternative permitted tighter experimental control during testing, and also because earlier work [14,16] suggested that this is a more-powerful technique for detecting effects of social partners on pituitary-adrenal activity than is the home-cage-separation procedure. The experiment was also designed to address certain potential shortcomings of existing work in this field. In previous studies of adults and juveniles reviewed above [2, 14, 17, 18] it has been customary to form social units about 1 month prior to the initiation of testing. It is possible that social groups need to be more long-standing before the presence/absence of a social partner is sufficient stimulus to affect pituitary-adrenal activity. Thus, in the present study, partners were drawn from stable groups in existence for a minimum of 6 months prior to testing.

Further, earlier studies have focused on plasma cortisol changes occurring within an hour of the experimental treatment [2, 14, 17, 18]. Because the effects on hormone concentrations may take longer to develop [4,16], the present experiment examined plasma cortisol levels at 1 and 25 hr following manipulation. The same earlier studies have assessed the influence of the presence or absence of familiar peers without taking into account the relative strengths of the relationship existing between individuals. One might expect that the presence/absence of an affiliative social companion would have a greater impact on pituitary-adrenal activity than would manipulation of an equally familiar, but less affiliative partner. Accordingly, the present study attempted to vary the affiliativeness of the familiar social companion. Behavioral measures of distress and inter-animal interaction were also collected for comparison with cortisol levels.

In sum, in the present study adult females from stable social groups were exposed to a novel environment either while alone or while with another female that was either familiar and judged to be affiliative with the subject, familiar but not judged to be affiliative, or unfamiliar. Blood samples for analysis of plasma cortisol levels were collected at 1 and 25 hr following novelty exposure. Behavioral measures of distress and of inter-animal interaction were recorded.

METHOD

Animals

Thirty nine, mother-reared, adult (>4 years of age) female squirrel monkeys (*Saimiri sciureus*) of the Colombian variety served as experimental animals. All females had been housed for at least 6 months immediately preceding the study in stable breeding groups consisting of 1 or 2 males and about 8–10 females.

Experimental Conditions

Seven or eight females were assigned to each of five experimental conditions: (1) Base (N=7); (2) Alone (N=8); (3) Familiar partner, judged to be affiliative (FA; N=8); (4) Familiar partner, not judged to be affiliative (FNA; N=8); and (5) Unfamiliar Partner (UNF; N=8). The variables of age and birth place (wild-born vs. laboratory-born) were approximately balanced across conditions.

The Base condition provided an estimate of resting plasma cortisol levels. Prior to any other disturbance, the subject was rapidly captured and a blood sample was collected. In each of the other four conditions, the experimental animal was removed from its home group and housed in a small test cage (74×64×38 cm) in an unfamiliar room for the succeeding 25 hr. Food and water were freely available and lighting was from 0700–1800 hr. In the Alone condition, only the subject was placed into the test cage. In the FA condition, two females from the same home group that had been judged to be an affiliative pair (see below) were placed together into the test cage. In the FNA condition, two females from the same home group that had not been judged to be an affiliative pair were placed together into the test cage. Finally in the UNF condition, one female from each of two separate (and visually isolated) groups were placed together into the test cage.

For the FA and FNA conditions, judgments of affiliation were made by a laboratory technician and an animal caretaker, both of whom had worked with the monkeys on a daily basis for several years and were familiar with individual animals. These judgments were based on estimates of spatial proximity both during waking hours and while the animals were asleep (i.e., which females slept next to each other) and on subjective assessments of positive social behavior. Most judgments were made during the course of the technician's daily rounds of the colony and of the caretaker's regular activities, although the technician spent an additional several hr observing the animals for this purpose. Pairs of females judged to be affiliative by at least one rater (provided the judgment was not disputed by the other rater) comprised the FA condition. Pairs of females that were observed, but were not judged to be affiliative by either rater were chosen as subjects for the FNA condition.

Behavior Observations

Several behaviors were monitored in the conditions involving exposure to novelty. Vocalizations and locomotor activity were observed because they previously have been found to reflect agitation during separation in juvenile squirrel monkeys [14,17]. These observations were made from 0 to 30 min and from 24 hr to 24 hr, 30 min following the beginning of testing. In each of these 30-min sessions, the subject was observed for 5, 2-min epochs, separated by 4-min intervals. For vocalizations, the simple frequency of occurrence was recorded. Because locomotor activity tends to occur in long temporal bouts, the number of 15 sec inter-

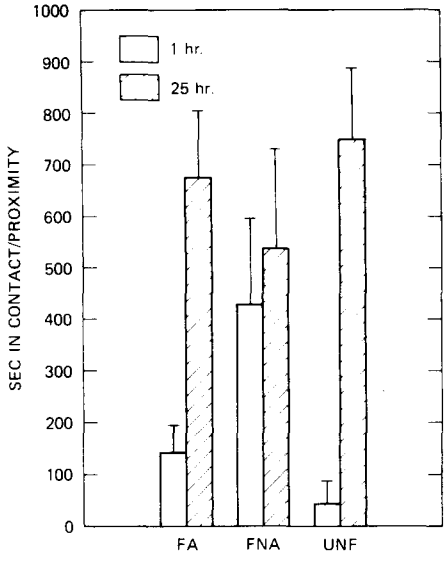


FIG. 1. Mean number of sec that females in the FA, FNA, and UNF conditions spent in physical contact or within 15 cm of each other. Observations were made for 15 min (900 sec) during the 1st and 25th hr of exposure to the novel room. Vertical bars represent standard errors of the means.

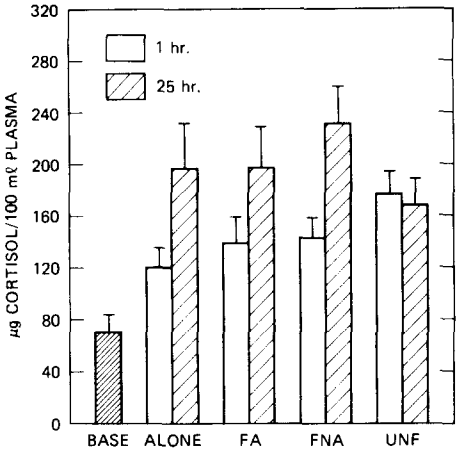


FIG. 2. Mean plasma cortisol levels of females in the five experimental conditions. In the four conditions involving exposure to novelty, cortisol levels following 1 and 25 hr of exposure were determined. Vertical bars represent standard errors of the means.

vals during which locomotion occurred was recorded. Additional observations in the FA, FNA, and UNF conditions were made during the 15 min immediately following observation of vocalizations and locomotor activity. A measure of inter-animal distance was obtained during this period by recording the total number of sec that the females were in contact/proximity (i.e., in contact or within 15 cm of each other, ignoring tails). Observations were also made of scent-marking activity and low-level agonistic behaviors (grab, active displace, passive displace, and genital display). However, these behaviors rarely occurred and will not be discussed further. All observations were made by experienced observers who were concealed in observation booths equipped with one-way glass.

Blood Sampling and Cortisol Determination

In the Alone, FA, FNA and UNF conditions, blood samples were collected 1 hr and 25 hr following placement into the novel room. For these samples, as well as for Base samples, the monkeys were rapidly anesthetized in a glass jar containing ether-soaked cotton. A blood sample (0.8 ml) was then taken via cardiac puncture using a heparinized injection needle (25 gauge×1.6 cm) and syringe. The blood from both animals (when 2 animals were tested) was collected within about 3–4 min of the initiation of the capture and anesthesia procedure. Thus, based on available evidence from laboratory rodents [7,28], effects of the capture and anesthesia procedures on circulating cortisol levels were minimal. All blood samples were collected between 0900 and 1100 hr. The blood was centrifuged to separate plasma, which was then extracted and frozen until assayed. Cortisol was determined in triplicate aliquots using a radioimmunoassay kit (^{3}H) from New England Nuclear.

RESULTS

The data from one pair of females in the UNF condition were discarded because the females were observed fighting during the first observation session. It was feared that their cortisol responses would reflect physical injury rather than the psychogenic factors under study.

Behavior

For Vocalizations, nonparametric tests were used because of heterogeneity of variance. Kruskal-Wallis analyses of variance by ranks revealed no differences among conditions during either the 1st or 25th hr of exposure. A Wilcoxon matched-pairs signed-ranks test showed that there was an overall decline in vocalizing from the 1st to the 25th hr ($\bar{X}$s=27.80, 1.03; $p<0.01$).

Locomotor activity was analyzed with a 4 (Conditions) ×2 (Sessions) analysis of variance (ANOVA), with the last factor treated as a repeated measure. The ANOVA yielded only a significant main effect for Sessions, $F(1,26)=15.79$, $p<0.01$. Locomotor activity decreased from the first to the second observation session ($\bar{X}$s=18.06, 8.47).

For contact/proximity, the pair was used as the unit of analysis. A 3 (Conditions—FA, FNA, UNF) ×2 (Sessions) ANOVA with the last factor treated as a repeated measure yielded a significant main effect for Sessions, $F(1,8)=35.52$, $p<0.01$, and a significant Conditions × Sessions interaction, $F(2,8)=5.49$, $p<0.05$. Inspection of Fig. 1 suggests that FNA animals tended to remain in contact/proximity somewhat more than did FA and UNF monkeys at 1 hr, and somewhat less than did animals of these two groups at 25 hr. However, further evaluation of the interaction with tests for simple main effects revealed no differences among conditions at

either 1 or 25 hr. Examining the interaction from a different perspective, tests for simple main effects showed that there was an increase in contact/proximity from 1 to 25 hr for the FA ($p<0.01$) and for the UNF ($p<0.01$), but not for the FNA conditions ($p>0.10$). Every pair of females in the FA and UNF conditions exhibited a greater increase in contact/proximity from the 1st to the 25th hr than did any pair of females in the FNA condition.

Cortisol

Dunnett's [9] procedure for multiple comparisons with a control was used to determine if a significant increase in plasma cortisol over the Base value occurred during the first hour of exposure to the novel room. Elevations were significant for the FA, FNA and UNF conditions, $t(32)=2.97$–4.29, $p<0.05$, but not for the Alone condition, $t(32)=2.18$, $p>0.10$; Fig. 2. Further analysis of cortisol levels in the four conditions involving exposure to novelty was performed with a 4 (Conditions) ×2 (Sessions) ANOVA with the last factor treated as a repeated measure. A significant main effect for Sessions, $F(1,26)=23.86$, $p<0.01$, and a significant Conditions × Sessions interaction, $F(3,26)=3.96$, $p<0.05$, were obtained. Examination of the interaction effect with tests for simple main effects revealed no differences among conditions at either 1 or 25 hr. However, when changes in cortisol levels from 1 to 25 hr were assessed, simple main effects showed that there was an increase in plasma cortisol concentrations with increasing time in the novel room for monkeys in the Alone ($p<0.01$), FA ($p<0.05$) and FNA ($p<0.01$) conditions, but not for animals in the UNF condition ($p>0.10$). Inspection of individual cortisol values showed that none of the 6 monkeys in the UNF condition exhibited an increase of 15 μg% or more from 1 to 25 hr, whereas 21 of 24 monkeys in the Alone, FA and FNA conditions did so. For 17 of these 24 monkeys, cortisol levels were more than 40 μg% greater at 25 hr than at 1 hr.

DISCUSSION

Plasma cortisol levels of females exposed to the novel room with a familiar partner were no lower than were those of females exposed to the room individually. In fact, it was only in the Alone condition that the elevation of plasma cortisol over the resting level was not significant at the end of 1 hr. Thus, among adult females, which appear to constitute the most cohesive subgroup of adult squirrel monkeys, the presence of a familiar companion did not reduce pituitary-adrenal activity after either 1 or 25 hr of exposure to novel surroundings. Moreover, this was the case despite the fact that females were from stable social groups, and in the FA condition, appeared to be particularly affiliative toward each other. In another recent experiment, separation of male-female pairs of *Saimiri* had no effect on plasma corticosteroid levels [26]. In all, studies of adult and juvenile males and females under various conditions of testing indicate that, in this species, the disruption of social relationships other than that between mother and infant has little effect on the activity of the pituitary-adrenal system.

As hypothesized above, only the disruption of especially intense social relationships may be sufficient stimulus to evoke increased pituitary-adrenal activity. It is certainly true that, in the squirrel monkey, the mother-infant bond constitutes a particularly intense or cohesive social relationship [19, 20, 29]. Complementary findings have been reported in another New World primate, the titi monkey (*Callicebus moloch*). Titi males and females form strong monogomous pair bonds and share in infant care-giving activities [8,11]. Preference tests have indicated that the male and female are more strongly attracted to each other than either is to their infant [25]. In separation tests, both parents exhibited plasma cortisol elevations when the adult partner was removed from the home cage, but neither showed this response when the infant was removed [25]. Thus, in both *Saimiri* and *Callicebus*, it is the disruption of a particularly intense social relationship that activates the pituitary-adrenal system, but the particular relationship involved varies between species.

In the present study, cortisol responses were identical in the FA and FNA conditions. Of course, one cannot be certain that the FA monkeys did, in fact, constitute a particularly affiliative subgroup of animals because assessments of this variable were based on subjective judgments. Therefore, the possibility that the presence of affiliative partners (as determined by some other procedure) could have moderated the cortisol response of our subjects cannot be ruled out at this time. However, the contact/proximity data indicate that the FA and FNA females did, at the least, comprise separate populations of animals since the FA but not the FNA females showed an increase in contact/proximity from the first to the second observation period. Behavioral measures of distress (vocalizations, locomotor activity) did not discriminate among conditions.

The one class of social partner that did affect the cortisol response to novelty was the unfamiliar female. Whereas cortisol levels increased from 1 to 25 hr in the Alone condition, there was no change from 1 to 25 hr in the UNF condition. As noted above, one pair of females in the UNF condition were observed to be fighting and were discarded from the study. It is possible that the other pairs of animals in this condition fought following the conclusion of the first hours of observation, and that this activity influenced cortisol levels at 25 hr. Rats that fight during exposure to electric shock show less pituitary-adrenal activity than do rats shocked but not allowed to fight [30]. It is conceivable that fighting which could have occurred after the first hr of testing prevented the UNF monkeys from showing a cortisol increase at 25 hr.

Previous studies have shown that various changes in neuroendocrine activity also follow the formation of all male and mixed-sex groups of adult *Saimiri* [2, 12, 24, 26]. It seems that in adult squirrel monkeys, neuroendocrine function is more responsive to the psychogenic processes inherent in the formation of new adult social relationships than to those accompanying the removal of established adult social companions.

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